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Systems and methods for real-time biopsy needle and target tissue visualization

Abstract

The present disclosure relates to the field of endoscopy. In particular, the present disclosure relates to systems and methods for real-time visualization of a target tissue, and which allows the location/orientation of the biopsy needle to be determined prior to its first actuation.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of Ser. No. 15/833,381, filed Dec. 6, 2017, which claims the benefit of priority under 35 U.S.C. § 119

FIELD

(1) The present disclosure relates to the field of endoscopy. In particular, the present disclosure relates to systems and methods which allow for real-time visualization of a target tissue, and which allow for efficient and accurate location/orientation of a biopsy needle prior to its first actuation and sample acquisition.

BACKGROUND

(2) Radial endobronchial ultrasound (R-EBUS) provides a minimally invasive option when clinical presentation indicates that tissue biopsy within the pulmonary passages is necessary. Conventional R-EBUS transbronchial needle aspiration (TBNA) involves delivering a radial ultrasound probe to the target airway through the working channel of a bronchoscope, visualizing the target pulmonary nodule on R-EBUS, locking placement of an access sheath, removing the radial ultrasound probe from the access sheath and then blindly advancing a biopsy needle to acquire cellular matter for cytologic evaluation. The inability to visualize the biopsy needle until after the tissue sampling procedure has been initiated often results in the biopsy needle completely missing the target nodule. To help ensure that the target nodule is successfully biopsied, the medical professional typically actuates the biopsy needle into the pulmonary tissue multiple times while rotating the bronchoscope. Such repetitive biopsy needle actuations may result in a variety of negative medical outcomes, including, unnecessary trauma to healthy tissues, excessive bleeding, pleural sac punctures (e.g., pneumothorax), blood vessel punctures, increased procedure duration and/or cost and potentially misdiagnosis (e.g., false-negatives).

(3) There may be a clinical advantage, particularly in the field of pulmonary endoscopy, for a tissue sampling system which allows a medical professional to visualize a biopsy needle and target tissue in real-time prior to the first needle actuation.

SUMMARY

(4) The present disclosure, in its various aspects, provides advantages in the medical field, such as the field of pulmonary endoscopy, for a sampling system that allows real-time visualization of pulmonary nodules, and which allows the location/orientation of the biopsy needle to be efficiently and accurately determined prior to its first actuation and sample acquisition.

(5) In one aspect, the present disclosure relates to a device comprising a first component that includes a proximal end, a distal end, and a lumen extending therebetween, and a tissue sampling element attached to the distal end of the first component. The proximal end of the first component may include a recessed portion. The tissue sampling element may include, e.g., a biopsy needle. The tissue sampling element may include a substantially linear configuration. The tissue sampling element may be moveable between a substantially linear configuration and a substantially curved configuration.

(6) In another aspect, the present disclosure relates to a system comprising a first component that includes a proximal end, a distal end, and a lumen extending therebetween, and a second component that includes a proximal end, a distal end, and a lumen extending therebetween. The proximal end of the first component may be removably attached to the distal end of the second component to form a contiguous lumen. An ultrasound catheter may extend through the contiguous lumen of the first and second components. An exterior tube may be slidably disposed about the first component, second component and ultrasound catheter. The ultrasound catheter may include an ultrasound probe slidably disposed within a sheath which forms an interference fit with the lumen of the second component. A tissue sampling element may be attached to the distal end of the first component. The proximal end of the first component may include a recessed portion configured to receive a post extending from a distal end of the second component. The post of the second component may form an interference fit with the recessed portion of the first component. The

lumen of the first component may align with the lumen of the second component to form a contiguous lumen when the post is disposed within the recessed portion. A portion of the sheath may extend distally beyond the ultrasound probe. A portion of the sheath may include a braided material. The braided material may extend along a proximal portion of the ultrasound probe. The portion of the sheath that extends distally beyond the ultrasound probe may include an unbraided material. The sheath may include a proximal end, a distal end and a lumen extending therebetween. The lumen of the sheath may include a first diameter portion and a second diameter portion. The system may further comprise a delivery device that includes a working channel configured to slidably receive the exterior tube. A proximal end of the ultrasound probe may be connected to a motor drive unit. The system may further include a delivery device comprising a working channel configured to slidably receive the exterior tube.

(7) In another aspect, the present disclosure relates to a method comprising advancing a tissue sampling system through a body passage, wherein the tissue sampling system includes interlocked first and second components removably disposed about an ultrasound catheter; imaging, with the ultrasound catheter, a target tissue with the body passage; advancing the tissue sampling system such that a portion of the first component penetrates the target tissue; and withdrawing the tissue sampling system from the body passage. The method may further comprise, prior to advancing the tissue sampling system, rotating the tissue sampling system to align the first component with the target tissue. The tissue sampling system may be advanced simultaneous with the imaging of the target tissue.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting examples of the present disclosure are described by way of example with reference to the accompanying figures, which are schematic and not intended to be drawn to scale. In the figures, each identical or nearly identical component illustrated is typically represented by a single numeral. For purposes of clarity, not every component is labeled in every figure, nor is every component of each embodiment of the disclosure shown where illustration is not necessary to allow those of skill in the art to understand the disclosure. In the figures:

(2) FIG. 1 provides a schematic view of a tissue sampling system, according to one embodiment of the present disclosure.

(3) FIG. 2 provides a schematic view of a distal portion of an ultrasound catheter, according to one embodiment of the present disclosure.

(4) FIG. 3 provides a schematic views of a connector component, according to one embodiment of the present disclosure.

(5) FIGS. 4A-4B provide schematic views of a sampling component that includes a straight (FIG. 4A) or curved (FIG. 4B) tissue acquisition element, according to one embodiment of the present disclosure.

(6) FIG. 5 provides an enlarged schematic view of a connector component and sampling component interlocked onto an ultrasound catheter, according to one embodiment of the present disclosure.

(7) FIG. 6 provides a schematic view of a distal portion of an exterior tube, according to one embodiment of the present disclosure.

(8) FIGS. 7A-7F illustrate the steps involved in sampling a pulmonary nodule, according to one embodiment of the present disclosure.

(9) FIGS. 8A-8C illustrate the steps involved in removing a tissue sample from a tissue acquisition element, according to one embodiment of the present disclosure.

(10) It is noted that the drawings are intended to depict only typical or exemplary embodiments of

the disclosure. Accordingly, the drawings should not be considered as limiting the scope of the disclosure. The disclosure will now be described in greater detail with reference to the accompanying drawings.

DETAILED DESCRIPTION

(11) Before the present disclosure is described in further detail, it is to be understood that the disclosure is not limited to the particular embodiments described, as such may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting beyond the scope of the appended claims. Unless defined otherwise, all technical terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure belongs. Finally, although embodiments of the present disclosure are described with specific reference to real-time visualization and sampling of pulmonary nodules, the systems and methods disclosed herein may be used to obtain biopsy samples from within a variety of body lumens, including, for example, the heart, vascular system, circulatory system, gastrointestinal (GI) tract, stomach, esophagus, urogenital system and the like.

(12) As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used herein, specify the presence of stated features, regions, steps elements and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components and/or groups thereof.

(13) As used herein, the term “distal” refers to the end farthest away from a medical professional when introducing a device into a patient, while the term “proximal” refers to the end closest to the medical professional when introducing a device into a patient.

(14) The present disclosure generally provides a tissue sampling system that includes a sampling component reversibly (e.g., removably) connected to an ultrasound catheter by a keyed or press-fit interaction with a connector component attached to an outer surface of the ultrasound catheter. The coupling geometry of the connector component and sampling component prevents rotation and/or translation of the sampling component relative to the ultrasound catheter and provides a fixed alignment which allows the medical professional to know in which quadrant of the radial ultrasound image the sampling component will appear relative to the target nodule prior to each tissue sampling step.

(15) Referring to FIG. 1, in one embodiment, the present disclosure provides a tissue sampling system **100** comprising a first component **110** (e.g., sampling component) and a second component **120** (e.g., connector component) disposed about an ultrasound catheter **130** (e.g., radial ultrasound catheter). The first component **110** may further include a tissue sampling element **118** (e.g., biopsy needle, fine aspiration needle, biopsy brush, etc.) extending from the distal end **124**. An exterior tube **150** may be slidably disposed about the ultrasound catheter **130** and first and second components **110**, **120**. The ultrasound catheter **130** may include an ultrasound transducer **136** disposed at the distal end **132** of an ultrasound probe **138** (e.g., radial ultrasound probe). The ultrasound transducer **136** and ultrasound probe **138** may be slidably disposed within a sheath **140** that includes a proximal end (not depicted), distal end **144** and a lumen **146** extending therebetween. A proximal end (not depicted) of the ultrasound probe **138** may be attached to a motor drive unit (MDU) configured to advance (e.g., move distally) and retract (e.g., move proximally) the ultrasound probe **138** and ultrasound transducer **136** within the fixed sheath **140**. The sheath **140** may be formed from a variety of materials that provide the requisite mechanical properties to navigate through narrow and tortuous body passages.

(16) In the various embodiments, the position of the first and second components **110**, **120** about the ultrasound catheter **130** may provide a distinct advantage over conventional tissue sampling systems due to the close proximity of the tissue sampling element **118** to the ultrasound transducer

136. Specifically, configurations such as depicted may allow the tissue sampling element **118** to be positioned in close proximity to the target tissue (e.g., within 5 mm or less) as the radial ultrasound image of the target tissue is being generated (e.g., in real-time). The close proximity to the target tissue allows that tissue sampling element **118** to be much shorter than conventional biopsy needles. For example, while a conventional pulmonary biopsy needle may include a length of 50 centimeters or more, the tissue sampling element of the present disclosure may include a length of less than 25 cm (e.g., 20 cm or less, 15 cm or less, 10 cm or less, 5 cm or less, 2.0 cm or less). The dramatically shorter length of the tissue sampling element **118** allows for lower production costs, and also allows more reliable and accurate sampling of the target tissue.

(17) As illustrated in FIG. 2, a portion of the sheath **140** extending proximally from the ultrasound transducer **136** may include a braided material **140a** (e.g., interwoven strands of flexible polymeric, carbon fiber, metallic and/or textile materials etc.) which provides enhanced stiffness (e.g., pushability) and torqueability to allow the ultrasound catheter **130** and the first and second components attached to the sheath **140** to be distally advanced and proximally retracted within and through the exterior tube by actuating (e.g., pushing and pulling) a proximal end (not depicted) of the sheath **140**. In addition, or alternatively, a portion of the sheath **140** extending distally beyond the ultrasound transducer **136** may include an unbraided material **140b** (e.g., clear plastic, silicone and/or rubber materials, etc.) that does not influence ultrasound image quality and target nodule visualization, while providing a conduit through which a suitable fluid (e.g., isotonic saline, etc.) may be intermittently flushed for consistent and reliable propagation of ultrasound energy. The unbraided material **140b** may also provide sufficient flexibility and/or deformability to bend or deflect during the tissue acquisition step such that the tissue sampling element is not impeded or obstructed from penetrating the target nodule.

(18) In one embodiment, a portion of the sheath **140** extending distally beyond the ultrasound transducer **136** may include a “strip” of hyperechoic (e.g., radiopaque) material, which appears as a dark portion (e.g., slice) on the radial ultrasound image. For example, the hyperechoic material may include a suitable powdered material (e.g., barium sulfate, etc.) mixed into the polymeric material(s) comprising the sheath **140** prior to the extrusion process. In addition, or alternatively, the hyperechoic material may include a thin strip of metallic material (e.g., copper, brass, stainless steel etc.) embedded within or otherwise adhered and/or affixed to a portion of the sheath **140** extending distally beyond the ultrasound transducer **136**. Because the first component **110** and sheath **140** are fixed to each other both axially and rotationally, and the orientation of the ultrasound catheter **130** is fixed rotationally with respect to the first component **110**, the location of the hyperechoic strip on the radial ultrasound image may allow a medical professional to identify the relative location of the tissue sampling element **118** even if the tissue sampling element is positioned behind (e.g., proximal to) the ultrasound transducer **136**. For example, the “strip” of hyperechoic material may be disposed on a portion of the sheath **140** that is directly opposite (e.g., offset 180 degrees) from the tissue sampling element. While visualizing the radial ultrasound image, the exterior tube **150** of the tissue sampling system **100** may be rotated in real-time to position the “strip” of hyperechoic material directly opposite the target nodule prior to distally advancing the ultrasound probe to deliver the tissue sampling element **118** into the target nodule.

(19) FIG. 3 provides an isolated schematic view of the second component **120** of FIG. 1. The second component **120** may include a proximal end **122**, distal end **124** and lumen **126** extending therebetween. The second component **120** may further include a post **128** (e.g., arm, tab, etc.) extending from the distal end **124**. FIG. 4A provides an isolated schematic view of the first component **110** of FIG. 1. The first component **110** may include a proximal end **112**, distal end **114** and lumen **116** extending therebetween. The proximal end **112** may include a recessed portion **113** (e.g., pocket etc.) configured to receive the post **128** of second component **120** in a keyed or press-fit manner to interlock the first and second components **110**, **120** together (FIG. 5). In some embodiments, recessed portion **113** may be a through lumen that extends from the proximal end of

the first component **110** to the proximal end of the sampling element **118** to provide a contiguous lumen for extracting the sample once acquired, as described further below. The first component **110** may further include a substantially straight tissue sampling element **118** (e.g., biopsy needle, fine aspiration needle, biopsy brush etc.) extending from the distal end **124** for sampling a concentric target nodule. Alternatively, as illustrated in FIG. **4B**, the first component **110** may include a tissue sampling element **118** configured to move from a straight to bent configuration when advanced distally beyond, and released from constraint within, the exterior tube for sampling an eccentric target nodule. The tissue sampling elements of FIGS. **4A** and **4B** may be embedded within the materials which form the first component, e.g., during the polymer co-extrusion process and/or secured using a suitable resin, glue or epoxy etc. Alternatively, the tissue sampling element **118** may be removable from the first component after tissue acquisition in order to remove the sample with another device such as described below.

(20) Referring to FIG. **5**, the second component **120** may be attached to a distal portion of the sheath **140** of the ultrasound catheter **130** by advancing (e.g., sliding) the ultrasound catheter through the lumen **126** of the second component **120** to form an interference fit between an outer surface of the sheath **140** and an inner surface of the lumen **126**. The interference fit between the second component **120** and the outer surface of sheath **140** may be sufficiently strong to prevent the second component from moving axially and/or rotationally along the ultrasound catheter **130** during a medical procedure, but sufficiently weak that a medical professional may move (e.g., reposition) or remove (e.g., detach or separate) the second component **120** by applying simultaneous twisting and pulling/pushing forces with one (or both) hands. Alternatively, the second component **120** may be permanently affixed to the sheath **140** of the ultrasound catheter **130** by a suitable weld, solder, braze, adhesive, glue and/or resin. Still referring to FIG. **5**, with the second component **120** firmly attached to the ultrasound catheter **130**, the first component **110** may be advanced over a distal portion of the ultrasound catheter **130** such that the recessed portion **113** of the first component **110** receives, and forms an interference fit (e.g., keyed fit, or press-fit) with, the post **128** of the second component **120**. The interference fit between the recessed portion **113** and post **128** may establish a reversible interlock which properly aligns and couples the first and second components **110**, **120** to prevent radial and/or axial movement of the first component **110** relative to sheath **140** of the ultrasound catheter **130** throughout the duration of the medical procedure. Rotational and/or axial movement of the first component **110** along and/or about the sheath **140** of the ultrasound catheter **130** may be further restricted by frictional forces between the outer surface of the sheath **140** and inner wall of the lumen **116**. As above, the combined interference fit(s) between the first component **110**, second component **120** and ultrasound catheter **130** (e.g., outer surface of sheath **140**) may be sufficiently strong to prevent the first component from moving axially and/or rotationally along the ultrasound catheter **130** during a medical procedure (e.g., within the patient), but sufficiently weak that a medical professional may remove or detach the first component **110** from the second component **120** and ultrasound catheter **130** by applying sufficient force (e.g., twisting and/or pulling) with one (or both) hands.

(21) Referring to FIG. **6**, in one embodiment, the exterior tube **150** may include a proximal end (not depicted), a distal end **154** and a variable-diameter lumen extending therebetween. For example, a distal portion of the exterior tube **150** may include a lumen with first diameter **156a** configured to receive and protect (e.g., fit over) the second component **120**, first component **110** and ultrasound catheter **130**. The remaining portion of the exterior tube **150** may include a second diameter **156b**, smaller than the first diameter **156a** and configured to slidably receive the ultrasound catheter. The smaller second diameter **156b** may provide increased wall thickness to the exterior tube **150** for improved pushability, steerability and resistance to bending and/or kinking. In addition, or alternatively, the smaller diameter second diameter **156b** may constrain the ultrasound catheter along its length to prevent excessive bending within the exterior tube as the tissue sampling system is advanced through narrow and tortuous body passages. The exterior tube **150**

may be formed from a variety of materials that provide the requisite mechanical properties (e.g., stiffness, pushability, flexibility, torqueability) to navigate through tortuous body passages without bending, kinking and/or breaking. The exterior tube **150** may further include one or more braided materials (e.g., interwoven strands of flexible polymeric, carbon fiber, metallic and/or textile materials etc.) to provide enhanced stiffness, torqueability and/or flexibility along all (or a portion of) the exterior tube. The exterior tube may include an outer diameter **158** configured to pass through the working channel of a conventional bronchoscope (e.g., approximately 2.0 mm to approximately 4.0 mm). The exterior tube **150** is in no way limited to a dual-diameter lumen, but may include a variety of lumen diameters, including, but not limited to single-diameter lumens, tapered-diameter lumens and the like.

(22) Referring to FIGS. 7A-7F, in use and by way of example, a bronchoscope **2** may be advanced through the trachea and into a bronchial passage in the vicinity of a target pulmonary nodule (FIG. 7A). Depending on the type of pulmonary nodule visualized through the bronchoscope (e.g., concentric or eccentric), an appropriate first component **110** (e.g., straight or curved, respectively) may be press-fitted onto the second component **120** already attached to the ultrasound catheter **130**. The ultrasound transducer **136** may be positioned distally beyond the tip of the tissue sampling element **118** by advancing the MDU, as discussed above. The tissue sampling system **100** may then be proximally retracted to a predetermined position, e.g., by proximally retracting the sheath **140**, to position the exterior tube **150** over the ultrasound catheter **130** (e.g., covering the distal end of the sheath **140** and ultrasound transducer **136**) and tissue sampling element **118** of the first component **110**. Enclosing the ultrasound transducer **136** and tissue sampling element **118** within the exterior tube **150** may protect the ultrasound transducer and prevent the tissue sampling element **118** from engaging the endoscope lumen and/or prematurely puncturing the lung tissue prior to identifying the target pulmonary nodule. The exterior tube **150** may then be advanced (e.g., threaded, pushed etc.) through and advanced distally beyond a working channel **4** of the bronchoscope **2** into the bronchial passage adjacent to the pulmonary nodule **8** (FIG. 7B). A proximal end (not depicted) of the sheath **140** (e.g., the portion that includes the braided material **140a**) may then be distally advanced to position the ultrasound transducer **136** beyond (e.g., outside) the distal end **154** of the exterior tube **150** and provide an ultrasound image of the pulmonary nodule **8** (FIG. 7C). With the location and orientation of the pulmonary nodule **8** determined, the ultrasound probe **138** may be proximally retracted through the fixed sheath **140** via the MDU to position the ultrasound transducer **136** slightly behind (e.g., proximal to) the tip of the tissue sampling element **118** (FIG. 7D), thereby allowing the tissue sampling element **118** (and pulmonary nodule **8**) to be visualized on the radial ultrasound image. The tissue sampling system **100** may be rotated as necessary to align the tissue sampling element **118** with the pulmonary nodule **8**. Alternatively, rather than proximally retracting the ultrasound probe to visualize the tissue sampling element, the location of the tissue sampling element **118** relative to the ultrasound transducer **136** may be determined by visualizing a “strip” of hyperechoic material integrally formed within a portion of the sheath **140** on the radial ultrasound image. As above, the tissue sampling system may be rotated as necessary to align the tissue sampling element **118** with the pulmonary nodule **8** based on the position of the “strip” of hyperechoic material. The tissue sampling element **118** may then be advanced into the pulmonary nodule **8** by actuating a proximal end (not depicted) of the sheath **140** such that the ultrasound catheter **130**, and attached first and second components **110**, **120**, move distally through the exterior tube **150** (FIG. 7E). The sheath **140** may be actuated (e.g., extended and retracted) as many times as necessary to obtain sufficient tissue sample within the tissue sampling element **118**. Referring to FIG. 7F, a tissue sample may be obtained from an eccentric pulmonary nodule using a tissue sampling element **118** that includes a curved or bent configuration, as depicted in **4B**. When the medical professional determines that the tissue sampling element **118** contains a sufficient amount of tissue sample for cytologic analysis, the proximal end (not depicted) of the sheath **140** may be proximally retracted to position the

ultrasound transducer **136** and tissue sampling element **118** within the exterior tube **150**. With the ultrasound transducer **136** and tissue sampling element **118** disposed (e.g., protected) within the exterior tube **150**, the exterior tube **150** may be proximally retracted to remove the tissue sampling system **100** from the body passage and through the working channel of the endoscope.

(23) Referring to FIG. **8A**, after the tissue sampling system **100** has been removed from the patient, the first component **110** may be separated from the keyed or pressed-fit interlock with the second component **120**. The tissue sampling element **118** may then be removed from the first component and attached to a corresponding fitting **80**, e.g., luer-lock, of an ejection system **70** (FIG. **8B**). For example, the ejection system **70** (e.g., stylet gun, syringe assembly etc.) may include a plunger **72** attached to an elongate stylet **74** which passes through a spring **76** housed within a chamber **78**. In one embodiment, the recessed portion **113** may extend through (not shown) the length of the first component **110** such that a lumen of the tissue sampling element **118** aligns and is coextensive with the recessed portion **113**. When the plunger **72** is depressed, the spring **76** moves to a compressed configuration within the chamber **78** and actuates the elongate stylet **74** to pass through the fitting **80** and lumen of the tissue sampling element **118** to eject the tissue sample **82** for cytologic analysis (FIG. **8C**). Alternatively, the depressing the plunger may force a pulse of compressed air through the fitting **80** and lumen of the tissue sampling element (rather than a stylet) to eject the tissue sample.

(24) The medical devices of the present disclosure are not limited to bronchoscopes, and may include a variety of medical devices for accessing body passageways, including, for example, catheters, ureteroscopes, duodenoscopes, colonoscopes, arthroscopes, cystoscopes, hysteroscopes, and the like. Alternatively, the tissue sampling system of the present disclosure may be positioned within the patient in the absence of an accompanying medical device.

(25) Various components of the tissue sampling system (e.g., first component **110**, second component **120**, sheath **140**, exterior tube **150**) and ejection system **70**, may be unitarily formed from suitable polymeric materials using extrusion (e.g., injection molding) and/or die-casting technologies, as are known in the art. Non-limiting examples of suitable materials may include polyolefins; polyamides (e.g., nylon, such as nylon 12, nylon 11, nylon 6/12, nylon 6, nylon 66); polyesters (e.g., polyethylene terephthalate (PET), polybutylene terephthalate (PBT), polyethylene naphthalate (PEN), polytrimethylene terephthalate (PTT)); polyethers; polyurethanes; polyvinyls; polyacrylics; fluoropolymers; copolymers and block copolymers thereof, such as block copolymers of polyether and polyamide (e.g., PEBAX®); and mixtures thereof. Ultraviolet curable polymers, such as polyimides and acrylic or methacrylic polymers and copolymers can also be used. Other examples of suitable polymers that can be used in balloons include polyethylenes, polyethylene ionomers, polyethylene copolymers, polyetheretherketone (PEEK), thermoplastic polyester elastomers (e.g., Hytrel®) and combinations thereof. In addition, or alternatively, any or all of these components may comprise metallic, ceramic or hardened plastic materials, as are known in the art.

(26) The size, shape and/or configuration of the various components are not limited to those depicted in the figures. For example, the first and second components **110**, **120**, are not necessarily limited to the depicted circular and/or or oblong shapes and/or openings.

(27) All of the devices and/or methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the devices and methods of this disclosure have been described in terms of preferred embodiments, it may be apparent to those of skill in the art that variations can be applied to the devices and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit and scope of the disclosure. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope and concept of the disclosure as defined by the appended claims.

Claims

1. An apparatus comprising: an ultrasound probe comprising an ultrasound transducer disposed at a distal end of the ultrasound probe; a tissue sampling element; an elongate member comprising: a first lumen extending along a longitudinal axis of the elongate member, the first lumen configured to receive the tissue sampling element, a sheath comprising a first portion having a first flexibility and a second portion distal to the first portion, wherein the second portion has a second flexibility greater than the first flexibility, and a second lumen extending along the longitudinal axis of the elongate member through the first portion and the second portion of the sheath, the second lumen configured to receive the ultrasound probe; and a strip of hyperechoic material disposed in the second portion and oriented along the longitudinal axis of the elongate member, wherein the first portion of the sheath is configured to cooperate with the second portion of the sheath to maintain the first lumen and the second lumen in a fixed radial position relative to each other to maintain the tissue sampling element and the ultrasound probe in a fixed radial position relative to each other, wherein the strip of hyperechoic material is disposed at a point on a circumference of the sheath to indicate a position of the tissue sampling element in an image generated by the ultrasound transducer, and wherein the point is offset radially from the first lumen and the tissue sampling element.
2. The apparatus of claim 1, wherein the second portion comprises a portion of a polymeric sheath.
3. The apparatus of claim 2, wherein the strip of hyperechoic material comprises a strip of metallic material.
4. The apparatus of claim 3, wherein the strip of metallic material comprises one or more of stainless steel, copper, and brass.
5. The apparatus of claim 1, wherein the second portion extends distally beyond a distal end of the ultrasound probe.
6. The apparatus of claim 1, wherein the second lumen is configured to have a liquid flushed through as the image is generated by the ultrasound transducer.
7. The apparatus of claim 1, wherein the first portion comprises interwoven strands of a metallic material.
8. The apparatus of claim 1, wherein the point is offset radially from the first lumen and the tissue sampling element 180 degrees.
9. An apparatus comprising: an ultrasound probe comprising an ultrasound transducer disposed at a distal end of the ultrasound probe; a tissue sampling element biased to a bent configuration; an elongate member comprising: a first lumen extending along a longitudinal axis of the elongate member, the first lumen configured to receive the tissue sampling element and to constrain the tissue sampling element to a straight configuration when the tissue sampling element is disposed within the first lumen, a sheath comprising a first portion having a first flexibility and a second portion distal to the first portion, wherein the second portion has a flexibility greater than the first flexibility and a second lumen extending along the longitudinal axis of the elongate member through the first portion and the second portion of the sheath, the second lumen configured to receive the ultrasound probe; and a strip of hyperechoic material disposed in the second portion and oriented along the longitudinal axis of the elongate member, wherein the first portion of the sheath is configured to cooperate with the second portion of the sheath to maintain the first lumen and the second lumen in a fixed radial position relative to each other to maintain the tissue sampling element and the ultrasound probe in a fixed radial position relative to each other, wherein the strip of hyperechoic material is disposed at a point on a circumference of the sheath to indicate a position of the tissue sampling element in an image generated by the ultrasound transducer, wherein the point is offset radially from the first lumen and the tissue sampling element, and wherein the first portion is configured to transmit torque to rotate the strip of hyperechoic material

when the elongate member is disposed within a body lumen.

10. The apparatus of claim 9, wherein the second portion comprises a portion of a polymeric sheath.

11. The apparatus of claim 10, wherein the strip of hyperechoic material comprises a strip of metallic material.

12. The apparatus of claim 11, wherein the strip of metallic material comprises one or more of stainless steel, copper, and brass.

13. The apparatus of claim 9, wherein the strip of hyperechoic material is radiopaque.

14. The apparatus of claim 9, wherein the second portion extends distally beyond a distal end of the ultrasound probe.

15. The apparatus of claim 9, wherein the second lumen is configured to have a liquid flushed through when the probe including the ultrasound transducer is disposed in the second lumen.

16. The apparatus of claim 9, wherein the first portion comprises interwoven strands of a metallic material.

17. The apparatus of claim 9, wherein the point is offset radially from the first lumen and the tissue sampling element 180 degrees.
